



: : Project Work on : :

**CRIMES AGAINST WOMEN IN INDIA :
STATISTICAL OVERVIEW, CAUSES AND
POSSIBLE SOLUTIONS**

: : Submitted by : :

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Course:- Statistical Methods 3

: : Under the guidance : :

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Statistics project (3rd sem)

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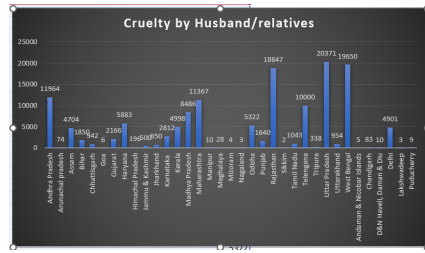
1 Abstract

In the ancient world women held a high place of respect in the society as mentioned in many scriptures. Volumes can be written about the status of our women and their heroic deeds from the vedic period to the modern times. But later on, because of social, political and economic changes, women lost their status and were relegated to the background. Many evil customs and traditions stepped in which enslaved the women and tied them to the boundaries of the house. Crime rates have seen a significant and concerning rise across the globe, with India being no exception. In this project, data about various types of crimes against women over time is collected and using the dataset, we can gain unparalleled insight into the prevalence and distribution of crimes against women over this period in different parts across India as well as within each state. Our main goal will be as follows :

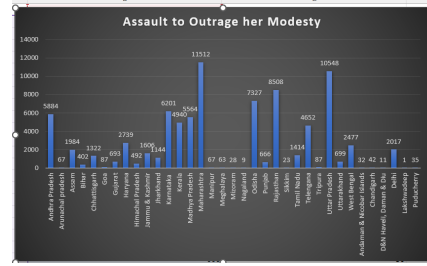
- i) To identify the states with the highest crime rates against women, and explore any potential correlations between crime statistics and social or economic factors in those states.
- ii) To analyze overall trends over time to assess whether government initiatives aimed at curbing crimes against women have been effective or not.
- iii) We are also interested to examine which type of crimes are most prevalent in each state/UT and come up with ways to reduce these crimes via policy measures or public outreach programs.

2 Introduction

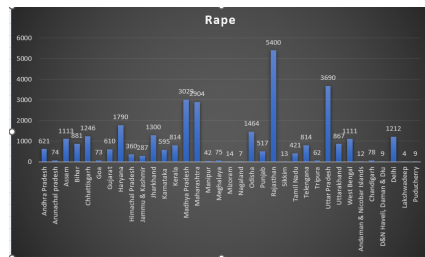
From the origin of social life in the country various centuries came and gone, time has changed people's mind and environment a lot, however violence against women is not seems to change a little bit. Time is the real eyewitness of all the sufferings (like sex discrimination, exploitation, oppression, aggression, degradation, humiliation, etc) bear by the helpless women. Women are so helpless in the Indian society where many female goddess are worshiped. In the Vedas, women are glorified as mother means one who can create and nourish a life. On



(a) Cruelty by husband



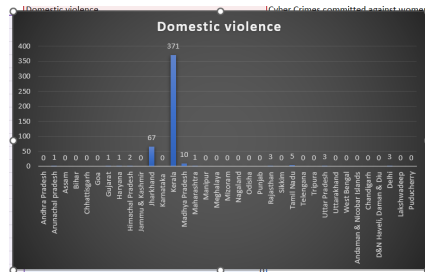
(b) Assault to outrage her modesty



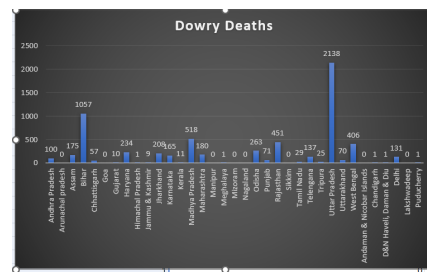
(a) Rape



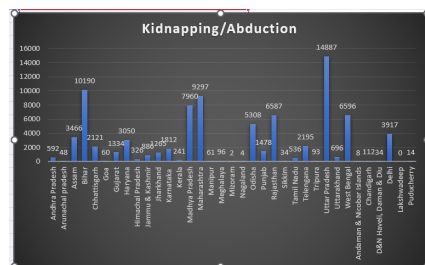
(b) Cyber crime committed against women



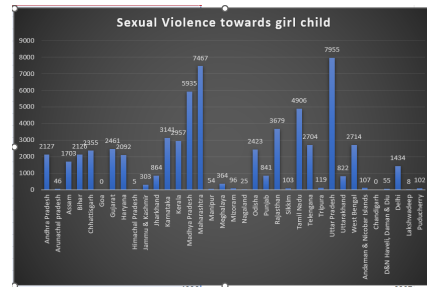
(a) Domestic Violence



(b) Dowry death



(a) Kidnapping/abduction



(b) Sexual violence towards women

So from the above figures we can find the states with highest number of crimes against women occurred in 2022 of different kind. We have it as follows :-

- a) **Trafficking - Maharashtra**
- b) **Insult to the modesty of women - Andhra Pradesh**
- c) **Acid Attack - West Bengal**
- d) **Abetment to suicide of women - Maharashtra**
- e) **Cruelty by husband - Uttar pradesh**
- f) **Assault to outrage her modesty - Maharashtra**
- g) **Rape - Rajasthan**
- h) **Cyber crime committed against women - Odisha**
- i) **Domestic Violence - Kerala**
- j) **Dowry death - Uttar pradesh**
- k) **Kidnapping/abduction - Uttar pradesh**
- l) **Sexual violence towards women - Uttar pradesh**

The above information may look horrible as Uttar pradesh becomes first in major crimes against women for 4 times and Maharashtra for 3 times ! We will do further analysis sometimes after , before that let us see the total crimes committed in every state in the year 2022.

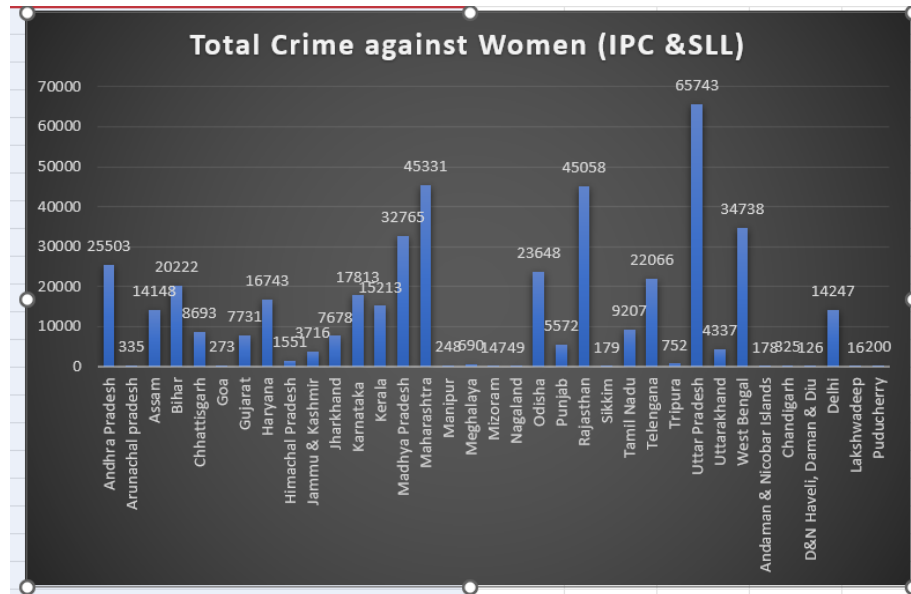


Figure 1: Total crimes

Here also we can see that Uttar Pradesh and Maharashtra hold the first and second position respectively ! Now let us look at the percentage of different types of crimes committed against women throughout the India in 2022 in the next figure -

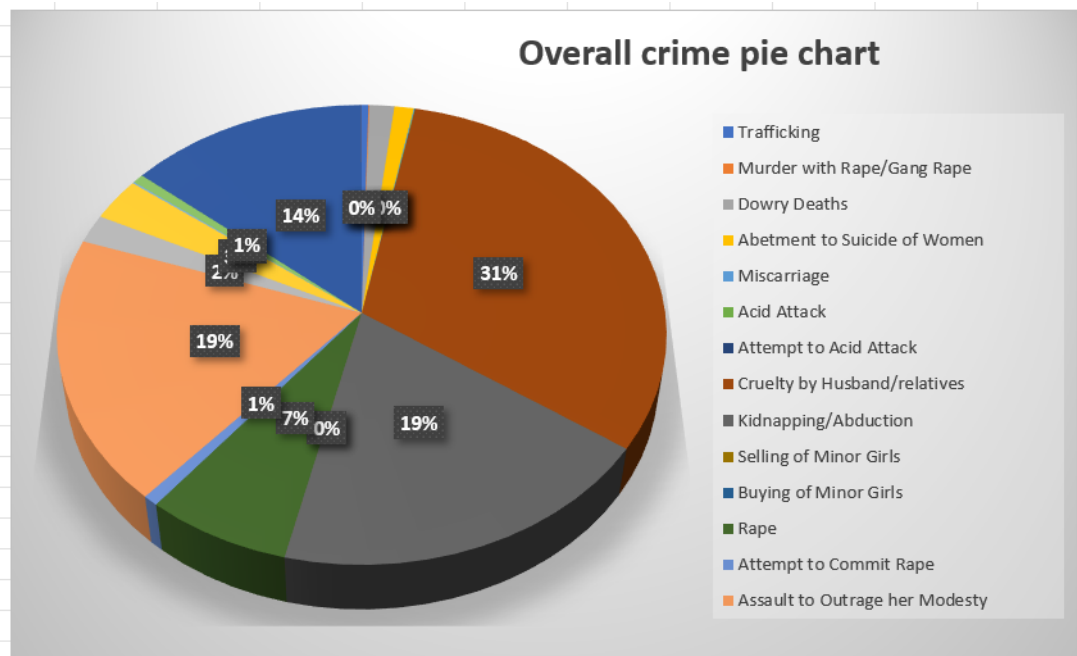


Figure 2: Overall crimes against women in India

So , 31% crimes occurred due to cruelty by husband or his family members . From here we may think that there must be some problem in socio - economic status in our country. So we are interested in looking into literacy rate and poverty rate of these states and then find some correlation between crime against women and socio-economic status.

Our primary observation is that literacy rate and poverty rate of a state has a significant impact on the crime rates. So first of all we have collected data about the poverty rate and literacy rate of each state of India. Then in the excel we make four columns namely "State/UT " , "Total Crime against Women (IPC &SLL)", "Literacy rate", "Poverty rate". Now we are using R to find out the partial correlation between number of crime committed and literacy rate given poverty rate. Then also find the partial correlation between number of crime committed and poverty rate given literacy rate.

The two above mentioned partial correlation computation in R is given in the next page...

3.1 R CODE

Partial correlation between literacy rate and number of crimes
committed given poverty rate

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/ALL.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 34 4

names(x)

## [1] "State.UT"
## [2] "Total.Crime.against.Women..IPC..SLL."
## [3] "Literacy.rate"
## [4] "Poverty.rate"

# Computing Partial correlation:-
f1 = lm(Total.Crime.against.Women..IPC..SLL.~Poverty.rate , data = x) # Regress Total.Crime.against.Wom
f2 = lm(Literacy.rate~Poverty.rate , data = x) # Regress Literacy.rate on Poverty.rate
f11 = f1$residuals # Taking residuals(true value - estimated value) of f1
f12 = f2$residuals # Taking residuals(true value - estimated value) of f2
r = cor(f11,f12) # Partial correlation between literacy rate and number of crimes committed
r # Print required partial correlation

## [1] -0.05489115
```

Partial correlation between poverty rate and number of crimes
committed given literacy rate

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/ALL.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 34 4

names(x)

## [1] "State.UT"
## [2] "Total.Crime.against.Women..IPC..SLL."
## [3] "Literacy.rate"
## [4] "Poverty.rate"

# Computing Partial correlation:-
f3 = lm(Total.Crime.against.Women..IPC..SLL.~Literacy.rate , data = x) # Regress Total.Crime.against.Wo
f4 = lm(Poverty.rate~Literacy.rate , data = x) # Regress Poverty.rate on Literacy.rate
f13 = f3$residuals # Taking residuals(true value - estimated value) of f3
f14 = f4$residuals # Taking residuals(true value - estimated value) of f4
r = cor(f13,f14) # Partial correlation between poverty rate and number of crimes committed
r # Print required partial correlation

## [1] 0.2116541
```

Also lets have a look into summary of f1 ,f2, f3, f4(defined in the code)

```
Call:
lm(formula = Total.Crime.against.Women..IPC..SLI. ~ Poverty.rate,
    data = x)

Residuals:
    Min       1Q   Median       3Q      Max
-22644  -9457  -6051   6539   47177

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8272.9      4200.0   1.970  0.0576 .
Poverty.rate  591.5       394.2   1.501  0.1432
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15800 on 32 degrees of freedom
Multiple R-squared:  0.06575, Adjusted R-squared:  0.03655
F-statistic: 2.252 on 1 and 32 DF, p-value: 0.1432
```

(a) Summary of f1

```
Call:
lm(formula = Literacy.rate ~ Poverty.rate, data = x)

Residuals:
    Min       1Q   Median       3Q      Max
-16.0680  -3.4225   0.2235   3.2465  12.0654

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  84.3502     1.7857   47.24 <2e-16 ***
Poverty.rate  -0.4492     0.1676   -2.68  0.0115 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.716 on 32 degrees of freedom
Multiple R-squared:  0.1833, Adjusted R-squared:  0.1578
F-statistic: 7.184 on 1 and 32 DF, p-value: 0.01153
```

(b) Summary of f2

```
Call:
lm(formula = Total.Crime.against.Women..IPC..SLI. ~ Literacy.rate,
    data = x)

Residuals:
    Min       1Q   Median       3Q      Max
-17521 -10718  -6024   5463   55050

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  41075.8     31059.0   1.321  0.196
Literacy.rate  -346.9       383.9  -0.904  0.373

Residual standard error: 16140 on 32 degrees of freedom
Multiple R-squared:  0.02488, Adjusted R-squared: -0.005591
F-statistic: 0.8165 on 1 and 32 DF, p-value: 0.3729
```

(a) Summary of f3

```
Call:
lm(formula = Poverty.rate ~ Literacy.rate, data = x)

Residuals:
    Min       1Q   Median       3Q      Max
-9.786  -3.849  -1.882   2.033  15.190

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  41.0774     12.3367   3.33  0.0022 **
Literacy.rate  -0.4082     0.1523   -2.68  0.0115 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.402 on 32 degrees of freedom
Multiple R-squared:  0.1833, Adjusted R-squared:  0.1578
F-statistic: 7.184 on 1 and 32 DF, p-value: 0.01153
```

(b) Summary of f4

So from the above R code we can see the partial correlation between number of crime committed and literacy rate given poverty rate is negative i.e if literacy rate decreased then number of crimes against women in a state or society will increase. Which is quite obvious. Again the partial correlation between number of crime committed and poverty rate given literacy rate is positive which means criminal offense will increase in a society if the poverty rate increased. Poverty forces a man to commit crime .So, **lack of education** and **increasing poverty rate** are main two factors of crimes against women.

Now just have a look of crime rate in India from 2015 to 2022. It is to be noted that this is not done with our dataset. It is an external dataset. We feel that it should be included here to aware people how crime rate is increasing day by day.

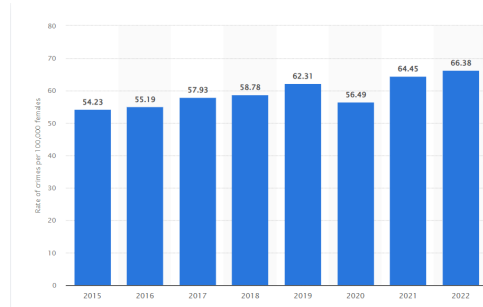


Figure 3: Trend of crime rate in India

4 Possible solutions (Government role)

1) **Pay special attention to the "hotspot" region** : - We have already marked Uttarpradesh and Maharashtra as "Hotspot" region of crimes against women. Government should immediately take serious action with these states like consulting with the local government , try to recruit more police there or to make strict laws against these type of crimes. Also they should give attention with those states where different types of crimes are at picked position.

2) **Economic empowerment**:- We have already seen how less literacy rate and high poverty rate affect the crime rate against women too much. Government should Provide women with financial literacy and vocational training to help them gain independence. Women who have their own income are less likely to stay in abusive relationships.

3) **Proper education system**:- Education is how much important that is cleared from the previous discussion . Government should launch more and more education programs and student scholarship to attract people towards education. Also government should sent representative to the door of the people to inform them about the value and importance of education

4) **Police reforms**:-Police can be reformed by including gender-focused recruitment and training, creating women police stations, and engaging women police volunteers in the service. Government have a very crucial role here.

5) **Judicial reforms**:- Establish fast track courts and increase punishment for serious cases like rape. Increase the representation of women in the judiciary. Government can make some women's quota in the judiciary system to give it a good strength.

6) **Listen to and believe survivors**:- When a woman shares her story of violence, she takes the first step to breaking the cycle of abuse. It is on all of us to give her the safe space she needs to speak up and be heard. It is important to remember that when discussing cases of sexual violence, a victim's sobriety, clothes, and sexuality are irrelevant. The perpetrator is the sole reason for assault and must bear the responsibility alone. Call out victim-blaming and counter the idea that it is on women to avoid situations that might be seen as "dangerous" by traditional standards.Survivors of violence are speaking out more than ever before, and everyone has a role to play to ensure they can have justice. Government should try to appoint health counselors at every district to give mental strength and support to that women.

7) **Teach the next generation and learn from them**:- The examples we set for the younger generation shape the way they think about gender, respect, and human rights. Start conversations about gender roles early on, and challenge the traditional features and characteristics assigned to men and women. Point out the stereotypes that children constantly encounter, whether in the media, on the street, or at school, and let them know that it is OK to be different. Encourage a culture of acceptance.

Talk about consent, bodily autonomy, and accountability in an age-appropriate way to boys and girls. For example, discuss the importance of a clear “yes” from all involved, the fact your body is yours and you make choices over what happens to it, and of how we must always take responsibility for our actions. It is important to also listen to what children have to say about their experience of the world. Government should teach this kind of education from school level to aware children about “good touch” and “bad touch”.Also , Values of respect and care can be inculcated in children from an early age. By empowering young advocates with information and educating them about women’s rights, we can build a better future for all.

8) Call for responses and services fit for purpose:- Services for survivors are essential services. This means that shelters, helplines, counselling, and all support for survivors of gender-based violence need to be available for those in need. Every year, the 16 Days of Activism campaign calls for united, global action to end all forms of violence against women and girls. Government must have 24 hours active helpline to prevent these cases also try to make a department for these purpose.

9) Fund women’s organizations :- Government should donate to local organizations that empower women, amplify their voices, support survivors, and promote acceptance of all gender identities and sexualities. Many women’s organizations worked to end violence against women, assist survivors, and secure equal rights for women and girls everywhere. Also government should create separate fund during annual budget to help these organization properly.

10) Media sensitivity :- The media can be sensitive to the plight of the rape victim and avoid highlighting the victim’s name or any inference that could lead to their identification. Government should control media strictly in such cases.

11) Self-defense:- Women can learn some kind of self-defense to protect themselves in difficult situations. Government can also help them in this case. Government should try to form self-defense academy for girls in each city.

12) Public Awareness:- The last but most important thing is to increase awareness of people. Public awareness campaigns can change societal attitudes towards women’s safety. Online movements like #MeToo and #CallItOut encourage women to speak out against violence and harassment. Government should encourage these types of programs more and more. Also they can organize public awareness campaign or street drama to aware people about these type of crimes.

5 Limitation of the project

Like every other project our project also have some limitations—

i) We have collected data from internet and the information about crime number , poverty rate , literacy rate etc are given by state government or local

government . So there is a high chance that some information may be wrongly given to us to hide the actual number.

ii) We are mainly concerned about socio - economic status of people which is the main cause of these crimes. It is also possible that some cases are happened not only due to lack of education or lack of money, wealth etc. It is possible that the victim may be mentally diseased person. Those information are missing in our data set.

iii) As census was not done in 2021 due to covid crisis, the data about literacy rate and poverty rate is calculated using population projection report. So , this is approximated and may be slightly different from the original figure.

6 Conclusion

Although several measures have been taken up by the Government of India to combat the crime against women, the crime rate is still increasing, which is quite alarming. The consequences of crimes against women extend beyond individuals and encompass the wider community and society to which these women belong. Probable challenges that impede the prevention of crime against women across the nation could be deep-rooted societal norms, persisting economic disparities, lack of empowerment, insufficient legal awareness, and inadequate implementation of laws. This is a complex issue, and therefore, one particular approach will not fit all the regions. Hence, region-specific interventions considering socio-cultural diversity within the region and demographic differences between the regions will be more effective in reducing the crime against women in the regions.

7 Acknowledgment

I would want to convey my heartfelt gratitude to Prof. Ayanendranath Basu my mentor, for his invaluable advice and assistance in completing my project. He was there to assist me every step of the way, and his motivation is what enabled me to accomplish my task effectively. I would also like to thank him for assisting me by supplying the equipment that was essential and vital, without which I would not have been able to perform efficiently on this project.

8 Bibliography

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